



## Research paper

## School based inclusive mentoring within the scope of an experiential learning model (IEM) for teacher education

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## ABSTRACT

This research employed school-based Inclusive Experimental Monitoring (IEM) mentoring within the scope of Kolb's (1984) experiential learning model. This was used to support pre-service teachers, combined with their existing undergraduate education at a Turkish university. The study undertook a mixed research approach, using the pre- and post-test experimental group (n=11) and control group (n=11). Various quantitative and qualitative data collection tools were used, including a personal information form, scales, questionnaires, video recordings, and unstructured observation. The results indicate that IEM has several positive outcomes for self-efficacy in inclusive education and readiness for the teaching profession. However, the study did not find any substantial changes in attitudes towards individuals with special educational needs, despite some observed positive impacts. It is believed that integrating IEM into the entire undergraduate education process can lead to further development in this regard.

## 1. Introduction

Inclusive education ensures that students with differing characteristics, including those with special educational needs, are taught in the same educational environment, regardless of their abilities and needs (UNICEF, 2017). This approach included in the education policies of many countries provides opportunities for all students (Ainscow et al., 2019). Throughout the globe, the number of students with special needs is currently increasing. In the USA, it increased from 6.5 million in 2009–2010 to 7.2 million during 2020–2021, with the majority receiving over 80% of their education in inclusive classrooms (National Center for Education Statistics, 2022). Furthermore, in China, the number increased from 239,000 in 2015 to 439,000 by 2020 (The State Council the People's Republic of China, 2021), while in Türkiye it increased by over 60% percent between 2016 and 2022 (Ministry of National Education [MNE], 2016; MNE, 2022).

This development has highlighted the importance of ensuring inclusive education forms part of teachers' training, including their professional development (Schuelka, 2018). This is particularly significant as successful inclusive classrooms require skilful planning and implementation (UNICEF, 2017). Furthermore, many students with special educational needs currently experience exclusion within an educational

environment (Strong, 2020), resulting in a negative impact on their emotional and mental health (Graham et al., 2019). Therefore, teachers require appropriate training to overcome these issues in inclusive classrooms.

## 1.1. Teacher training in inclusive education

The education of pre-service teachers is vital, as it forms the foundation for their professional lives. This has led to a number of studies focusing on undergraduate curriculums (e.g. Kaplan & Lewis, 2013). In addition, many countries have updated their policies concerning inclusive education within teacher training programmes (UNESCO, 2020). However, there remain a number of deficiencies in many countries, including the East Asia and the Pacific, where, as noted by UNICEF (2020), teacher training remains insufficient to meet the requirements of inclusive education.

There are a number of educational approaches to eliminating these inadequacies, including experiential learning (Girvan et al., 2016; Habib et al., 2021). This has led to a recognition of the importance of ensuring teacher training includes experience-based learning alongside theoretical knowledge (Kaplan & Lewis, 2013). However, while a number of studies have demonstrated the benefits (e.g. improving teachers'

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attitudes towards inclusive education and accepting students with special educational needs) of experiential learning processes for training teachers to teach those with special educational needs (e.g. Hill & MacDonald, 2016; Kang & Martin, 2018), there remains a limited degree of experimental research.

In addition, it is vital to examine the place of experiential learning in the teacher-training curriculum, particularly as many countries include school-based teaching practice as a key element (e.g. European Agency for Development in Special Needs Education, 2011). For example, the teaching practice directive in Türkiye (MNE, 2021) states that, in their final years of their undergraduate education, pre-service teachers are required to practice in schools for 6 h a week over a twelve-week period each semester (i.e. a total of 72 h). Pre-service teachers are also subject to be mentored by both a supervisor and the classroom teacher (MNE, 2021). However, Aslan and Sağlam (2018) noted that although this process is beneficial, it needs some improvement (European Agency for Special Needs and Inclusive Education, 2016).

### 1.2. Kolb's learning model

Kolb's (1984) learning model is considered to have a positive impact on inclusivity mentoring for pre-service teachers. It consists of four stages (Kolb, 1984; Lee, 2019). Firstly, the learner encounters an experience (i.e. Concrete Experience). Secondly (i.e. Reflective Observation), the learner is focused on thinking about this experience, including critically examining his/her own experiences and any issues that may arise. During the third stage (i.e. Abstract Conceptualisation), the learner focuses on introducing new theories to resolve problems or change existing ones. In the final stage (Active Experimentation), the learner uses the newly created or modified ideas or theories to solve problems in a new practical experience, including returning to the first stage of active experience to continue the cycle.

The experiential learning model is considered a powerful learning method in recognition that individuals need to understand themselves as learners and take responsibility for their learning processes (Institute of Experiential Learning, 2021). Within this framework, inclusive mentoring focuses on students with special educational needs within a classroom and has made a number of important contributions to the training of teachers (e.g. improving problem solving skills, professional satisfaction and feelings of high self-confidence) (Peterson, 2021; Whatman, 2021). However, mentoring for teaching experience with students with special educational needs may require a separate proficiency (Author, manuscript).

### 1.3. Research aim and questions

This research aims to integrate mentoring into the undergraduate curriculum by examining the impact of inclusive mentoring within the framework of teaching practice, based on Kolb's experiential learning model. In order to make this contribution, the study explores how IEM influences the development of pre-service teachers in inclusive education, focusing on self-efficacy—their confidence in their abilities to influence their motivations and decisions (Bandura, 1977), which is important for teacher development (Mintz et al., 2020). Additionally, research has underscored the importance of positive teacher attitudes towards students with special educational needs in fostering high-quality inclusive environments over the years (e.g., Forlin et al., 2007; Kunz et al., 2021). Therefore, another objective of this study is to understand how IEM shapes pre-service teachers' attitudes. Finally, as another significant element, the study examines how IEM contributes to teacher readiness, defined as educators' competence and preparedness to effectively teach in the classroom (Inan & Lowther, 2010). Thus, the research addresses the following questions:

- i) How does the School-Based Inclusive Mentoring Approach, within the context of the Experiential Learning Model, influence the self-efficacy of pre-service teachers in inclusive education?
- ii) What impact does the School-Based Inclusive Mentoring Approach, within the framework of the Experiential Learning Model, have on attitudes towards children with special educational needs?
- iii) What role does the School-Based Inclusive Mentoring Approach play within the framework of the Experiential Learning Model in preparing pre-service teachers for the teaching profession?

## 2. Method

### 2.1. Research design

To address the research questions stated above, the researcher adopted a nested mixed design within the framework of a mixed research approach (Creswell & Creswell, 2017). This allows one type of data (quantitative or qualitative) to be embedded within another, permitting the collection of different forms of qualitative and quantitative data, i.e. simultaneous, sequential, simultaneous and sequential (Cresswell & Plano Clark, 2011).

The current research employed a quasi-experimental design, in the form of a pre-test–post-test approach with a non-equivalent control group (Creswell & Creswell, 2017). This enabled the selection of appropriate volunteers within the parameters of inclusive education. The data was used in a qualitative context so as to ensure depth in the quantitative data.

### 2.2. Participants

The participants were selected using criterion sampling (Patton, 2015). Accordingly, being a final-year pre-service teacher from the primary teaching department at a Turkish university was determined as a prerequisite. These were selected due to the importance of primary education (UNICEF, 2023), as well as primary school teachers being a key factor in inclusive education (Shevchenko et al., 2020). In addition, the school-based course and the compulsory course in special education and inclusion are only available at this level of the undergraduate teaching curriculum.

The researcher invited sixty-four final-year pre-service teachers from a state university, who participated in teaching practice as a compulsory element of their undergraduate education. Twenty-three were found to have students with special educational needs in their classes. Of the remainder, twenty-one stated that the teacher in their classroom believed their student/s had special needs, but their families were reluctant to obtain an official diagnosis.

As detailed in Table 1, eleven of the pre-service teachers volunteered to participate in this study. Four were undertaking their teaching practice in classrooms with students diagnosed with learning disabilities, which typically manifest in reading, writing, and math, while the remaining seven were in classrooms with students with attention deficit and hyperactivity.

This research also used a comparable control group, consisting of eleven participants with characteristics approximating to those in the experimental group. Seven were undertaking their teaching practice in classrooms with students diagnosed with learning disabilities, while the remainder were with students with attention deficit and hyperactivity.

While the experimental group was made up of nine females and two males (P4 and P8), the control consisted of eight females and three males. The mean age was 21.7 in the experimental group and 21.5 in the control, while both groups had an academic grade average of between 3 and 3.5 out of 4. None of the volunteers had previously taken a course on inclusion directly, and none (apart for one participant in the experimental group (P6)) had any direct experience of special needs, including in their family or immediate environment. Both groups confirmed that

**Table 1**  
Characteristics of the participants.

| Characteristics of Experimental Group Participants |     |        |                        |                   | Characteristics of Control Group Participants |     |        |                        |                   |
|----------------------------------------------------|-----|--------|------------------------|-------------------|-----------------------------------------------|-----|--------|------------------------|-------------------|
| Participants                                       | Age | Gender | Academic Grade Average | Pupils' diagnosis | Participants                                  | Age | Gender | Academic Grade Average | Pupils' diagnosis |
| P1                                                 | 21  | F      | 3.00–3.50              | AHDH              | C1                                            | 21  | F      | 3.00–3.50              | ADHD              |
| P2                                                 | 21  | F      | 3.00–3.50              | AHDH              | C2                                            | 21  | F      | 3.00–3.50              | AHDH              |
| P3                                                 | 22  | M      | 3.00–3.50              | LD                | C3                                            | 22  | M      | 3.00–3.50              | LD                |
| P4                                                 | 22  | F      | 3.00–3.50              | AHDH              | C4                                            | 21  | F      | 3.00–3.50              | ADHD              |
| P5                                                 | 21  | F      | 3.00–3.50              | LD                | C5                                            | 21  | F      | 3.00–3.50              | LD                |
| P6                                                 | 21  | F      | 3.00–3.50              | AHDH              | C6                                            | 21  | F      | 3.00–3.50              | ADHD              |
| P7                                                 | 24  | M      | 3.00–3.50              | AHDH              | C7                                            | 23  | F      | 3.00–3.50              | LD                |
| P8                                                 | 23  | F      | 3.00–3.50              | AHDH              | C8                                            | 23  | M      | 3.00–3.50              | LD                |
| P9                                                 | 21  | F      | 3.00–3.50              | AHDH              | C9                                            | 21  | F      | 3.00–3.50              | LD                |
| P10                                                | 22  | F      | 3.00–3.50              | LD                | C10                                           | 22  | F      | 3.00–3.50              | LD                |
| P11                                                | 21  | F      | 3.00–3.50              | LD                | C11                                           | 21  | M      | 3.00–3.50              | LD                |

P: Participant code in the experimental group, C: Participant code in the control group, F: Female, M: Male, LD: Learning Disability, ADHD: Attention Deficit Hyperactivity Disorder.

they had been willing to enrol in the department of primary teaching and that they were happy to be there.

### 2.3. Research instruments

**Personal Information Form:** During the research process, a range of information was gathered from the participants, using a form created with the assistance of experts in the fields of special education, primary education, and grammar. The questions elicited the volunteers' age, gender, academic averages, and any courses they had followed related to inclusive education, alongside their personal experience of those with special educational needs. In addition, they were asked about their attitude to joining the department. The answers determined the demographic characteristics of the volunteers as well as which of them would form the control group.

**Scale 1—Scale for Self-Efficacy of Teacher Training Students in Inclusive Education Environments:** This scale was developed by Teke and Sözbilir (2021), whose participants consisted of 650 pre-service teachers. They collected data from 313 volunteers during the exploratory factor analysis process and from 337 volunteers during the confirmatory factor analysis process. In addition, expert opinion was used for content validity. The scale has seventeen items on a 5-point Likert type. The items in the scale are graded as strongly agree, agree, undecided, disagree, and strongly disagree. The scale has a two-factor structure, which is identified as a *cooperation* and *teaching process*. The Cronbach's alpha coefficients of the two factors of the scale and the whole scale were .904, .748 and .901, respectively. It was stated that the item-total correlation values of the scale ranged from .344 to .728. In the context of confirmatory factor analysis, it was stated that the model had acceptable fit values.

**Scale 2—Attitude Scale Towards Individuals with Special Needs:** This scale was developed by Yarah (2015), with data collected from 180 pre-service teachers studying in various departments using exploratory and confirmatory factor analysis. The scale consists of thirteen items and three factors, graded as strongly agree, agree, undecided, disagree and strongly disagree. The Cronbach's alpha coefficients of the sub-factors are *liking*, *sensitivity* and *negative attitude* and the scale consists of .803, .812, .712 and .841, respectively. The load values of the factors vary between .551 and .860, and the scale explains 58.592% of the total variance. It indicates that the item-total correlation coefficients range between .329 and .627. By considering the top and bottom 27% of the groups, an unrelated *t*-test was used to reveal that the scale items had a distinctive feature in terms of the attitudes of the pre-service teachers. The Spearman-Brown correlation value and the Guttman split-half value were calculated within the scope of the two-half test reliability, and these values were reported to be .772 and .751, respectively.

**Scale 3—Readiness Scale for Teaching Profession:** This scale was

developed by Yıldırım and Köklükaya (2017) and consists of a 5-point Likert scale of thirty items, graded as strongly agree, agree, undecided, disagree and strongly disagree. Expert opinion was used for content validity of the scale, and factor analysis was used for construct validity. The research involved 283 pre-service teachers from different branches. It found that the Kaiser-Meyer Olkin value was .95, factor loads ranged from .45 to .81, and the scale had a two-factor structure and explained 48% of the total variance. The Cronbach's alpha coefficients for the subscales and the whole scale are .95, .78 and .93, respectively. The lowest score that can be obtained from the scale is 30, and the highest score is 150.

**Questionnaires:** Two questionnaires were created for this research, being completed with the assistance of experts in special education, primary education, and grammar and piloted with ten pre-service teachers who did not participate in the research.

**Questionnaire-1.** Questionnaire-1 was applied to all participants both before and after the mentoring. It started with three closed-ended questions related to the volunteers' self-evaluation of their self-efficacy in inclusive education, their attitudes towards individuals with special educational needs, and their readiness for the teaching profession. The questions were graded from 1 to 5, as very inadequate, insufficient, undecided, sufficient and very adequate. These were asked again following the mentoring, followed by an open-ended question to obtain opinions about the undergraduate courses directly related to inclusion.

**Questionnaire-2.** At the end of the research, Questionnaire-2, consisting of five open-ended questions, was given to the experimental group, to obtain their opinions of inclusive mentoring, along with any suggestions for improvement.

**Video Recording:** The experimental group experienced the first and fourth stages of Kolb's experiential learning model in schools. In the second and third stages of Kolb's experiential learning model, reflective meetings were held between the participants and mentors through distance education platforms. Interactions between the mentors and pre-service teachers during the meetings were recorded for coding purposes. Thus, the aim was to obtain deep data focused on the experiential learning processes of the participants by using the videos included in the qualitative data collection tools (Creswell, 2013).

**Unstructured Observation:** The observation included seven online meetings, in addition to two face-to-face meetings (the first and last meetings) with the experimental group. Unstructured observation was used to collect holistic, comprehensive, and detailed data concerning the implementation process of the research (Given, 2008), being carried out by two researchers, who undertook the mentoring process and took weekly notes.

## 2.4. Research process

### 2.4.1. Creating inclusive experiential mentoring (IEM) for teacher education

During the first stage, the pre-teachers applied the lesson plans they had prepared in inclusive classrooms with students with special educational needs. In the second and third stages, the aim was to identify the problems they encountered and find solutions based on their experiences and consultancy with IEM experts. In the final stage, they were required to put their new knowledge into practice and to share the new outcomes with the experts. This process then continued in a cycle that repeated every week.

### 2.4.2. Application of IEM and the data collection process

IEM was applied in this research after the relevant ethical permissions had been obtained and the volunteers had signed the consent forms. The participants were then divided into the experimental and control groups. Both groups participated in the twelve-week school-based teaching practice, mentored by a supervisor from their faculty and the classroom teacher (MNE, 2021). All participants had 6 h of school-based practice and weekly 2-h theoretical meeting with their faculty supervisor. In addition, those in the experimental group participated in the IEM, after allowing a week for the volunteers to adapt to both the school and the students. The mentoring was introduced during the second week, and pre-data was collected face-to-face. IEM was commenced online during the third week. These became weekly meetings lasting approximately 2 h, with a 10-min break. The IEM was terminated after seven weeks, in accordance with a lessening of the pre-service teachers' issues and cyclical saturation of the data. A face-to-face meeting was held during the following week, which included the collection of the post-data.

## 2.5. Data analysis

The analysis of the quantitative data employed descriptive statistics and inferential statistics, including multivariate analysis (Çokluk et al., 2012). Firstly, the analysis confirmed that there was no missing data. Furthermore, Mahalanobis distances (Tabachnick & Fidell, 2013) revealed no extreme values with a  $p$  value less than .001. The histogram graph in the analysis of the distribution of the data was performed, with the skewness and kurtosis values converted into  $z$ -scores, and the Shapiro-Wilks (S-W) test (Taspinar, 2017). This revealed that the data had a normal distribution. In addition, scatter plots were used to confirm the normality and linearity of the data (Mertler & Vannatta, 2005). Independent sample  $t$ -test was utilised for the pre-test scores of both groups, showing parametric assumptions (normality, covariance) were met, and there was no significant difference in terms of scales and subscales,  $p > .05$ . This therefore evidences the equivalence of the experimental and control groups. Mixed M(A)NOVA was performed to evaluate the pre-and post-test measurements of the experimental and control groups, including assumptions of linearity between dependent variables, multivariate normality, and homogeneity of variance-covariance matrices of groups. Due to the equal number of participants in the experimental and control groups, repetitive M(A)NOVA was found to be robust (Leech et al., 2008). Mixed M(A)NOVA analysis (Leech et al., 2008) was carried out through the SPSS programme. The table in the source was used, based on Cohen (1988) in the interpretation of the effect value (Leech et al., 2008).

Thematic coding was used to analyse video recordings and qualitative data from Questionnaire-1 and Questionnaire-2 (Robson, 2011). The qualitative analysis followed a structured approach. First, written and visual data were reviewed and coded into text files. Themes were identified based on these codes, and findings were reported with each participant identified by a unique code (P1, P2, P3, etc.). In this process, the constant comparison method, pioneered by Glaser and Strauss (1967) and developed further by Strauss and Corbin (1998), played a

crucial role. This method involved continuously comparing data to uncover themes (Strauss & Corbin, 1998). Raw data was collected, preliminary codes were generated, and final codes were established using techniques like open, axial, and selective coding (Saldana, 2013; Strauss & Corbin, 1998).

During the analysis, insights from Saldana (2013) and Ekiz (2009) were strategically applied, with Excel used for data management. Questionnaire 1 consisted of three closed-ended questions with numerical responses, following principles outlined by Saldana (2013) for quantifying qualitative data. Questionnaire 2 included more detailed inquiries analysed inductively as per Ekiz (2009), with data organized in Excel for comparative analysis. Multiple readings informed preliminary and open coding, revealing connections and leading to the formation of sub-themes through axial coding. Selective coding involved higher-level analytical processes and team discussions (Saldana, 2013). Open-ended questions from both questionnaires underwent thematic coding and constant comparison, with researchers collaboratively finalizing codes after reviewing participant suggestions.

Direct quotes also supported the qualitative analysis, while unstructured observations enhanced the depth of the study. A descriptive qualitative analysis approach (Ekiz, 2009) ensured data integrity and minimized researcher bias (Miles & Huberman, 1994).

## 3. Results

The results are structured and presented within the context of the research problems, and they are reported thematically.

### 3.1. Theme 1: development of pre-service teachers' self-efficacy towards inclusive education

The study examined the impact of IEM on the participants' self-efficacy levels towards inclusive education through the total score obtained from Scale 1 and the total scores obtained from the sub-dimensions (cooperation with the teaching process) (see Table 2).

Table 2 demonstrates that the average post-test scores obtained from Scale 1 and its sub-dimensions were higher in the experimental group. The findings of the mixed M(A)NOVA analysis according to the values obtained from Scale 1 and its sub-dimensions are given in Table 3.

This confirms that evidence for significant multivariate effects was obtained based on group and time interactions, with the linear combinations of the related dependent variables containing evidence of differences between the experimental and control groups in the context of pre- and post-test. In addition, the effect value related to the interaction

**Table 2**

Descriptive statistics values according to the sub-dimensions of Scale 1 and Scale 1.

|                                                                             | Group        | Mean   | Std. Deviation | N  |
|-----------------------------------------------------------------------------|--------------|--------|----------------|----|
| Pre-test: Scale 1 total score                                               | Experimental | 70.364 | 6.697          | 11 |
|                                                                             | Control      | 72.181 | 7.613          | 11 |
|                                                                             | Total        | 71.273 | 7.059          | 22 |
| Post-test: Scale 1 total score                                              | Experimental | 75.546 | 6.948          | 11 |
|                                                                             | Control      | 71.364 | 6.021          | 11 |
|                                                                             | Total        | 73.455 | 6.696          | 22 |
| Pre-test: The total score of the teaching process sub-dimension of Scale 1  | Experimental | 19.818 | 2.183          | 11 |
|                                                                             | Control      | 21.182 | 2.359          | 11 |
|                                                                             | Total        | 20.500 | 2.325          | 22 |
| Post-test: The total score of the teaching process sub-dimension of Scale 1 | Experimental | 22.546 | 2.207          | 11 |
|                                                                             | Control      | 21.000 | 2.236          | 11 |
|                                                                             | Total        | 21.773 | 2.308          | 22 |
| Pre-test: The total score of the cooperation sub-dimension of Scale 1       | Experimental | 50.546 | 4.865          | 11 |
|                                                                             | Control      | 51.000 | 5.950          | 11 |
|                                                                             | Total        | 50.773 | 5.309          | 22 |
| Post-test: The total score of the cooperation sub-dimension of Scale 1      | Experimental | 53.000 | 5.040          | 11 |
|                                                                             | Control      | 50.364 | 3.957          | 11 |
|                                                                             | Total        | 51.682 | 4.623          | 22 |



**Table 3**  
Findings on Mixed M(A)NOVA analysis in the context of Scale 1.

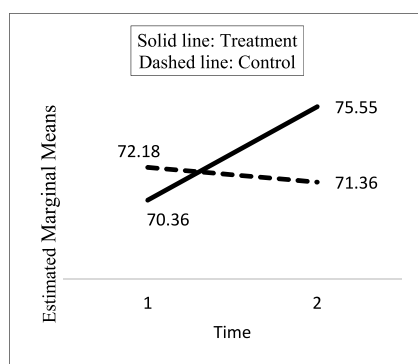
| Source                                    | SS     | df | MS     | F      | p    | $\eta_p^2$ | Power |
|-------------------------------------------|--------|----|--------|--------|------|------------|-------|
| Between-Subjects                          |        |    |        |        |      |            |       |
| Groups                                    | 15.364 | 1  | 15.364 | .177   | .679 | .009       | .069  |
| Within-Subjects                           |        |    |        |        |      |            |       |
| Time                                      |        |    |        |        |      |            |       |
| Scale 1                                   | 52.364 | 1  | 52.364 | 7.896  | .011 | .283       | .762  |
| Time*Group                                | 99.000 | 1  | 99.000 | 14.928 | .001 | .427       | .957  |
| Between-Subjects                          |        |    |        |        |      |            |       |
| Groups                                    |        |    |        |        |      |            |       |
| Teaching process sub-dimension of Scale 1 | .091   | 1  | .091   | .011   | .917 | .001       | .051  |
| Collaboration sub-dimension of Scale 1    | 13.091 | 1  | 13.091 | .290   | .596 | .014       | .081  |
| Within-Subjects                           |        |    |        |        |      |            |       |
| Time                                      |        |    |        |        |      |            |       |
| Teaching process sub-dimension of Scale 1 | 17.818 | 1  | 17.818 | 9.400  | .006 | .320       | .830  |
| Collaboration sub-dimension of Scale 1    | 9.091  | 1  | 9.091  | 1.862  | .188 | .085       | .255  |
| Time*Group                                |        |    |        |        |      |            |       |
| Teaching process sub-dimension of Scale 1 | 23.273 | 1  | 23.273 | 12.278 | .002 | .380       | .915  |
| Collaboration sub-dimension of Scale 1    | 26.273 | 1  | 26.273 | 5.382  | .031 | .212       | .598  |

Note. SS: Sum of Squares, MS: Mean Square.

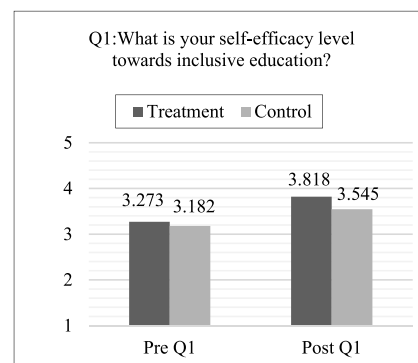
effect is large. The within the subject values regarding Scale 1's sub-dimension of teaching process show a significant difference in favour of the experimental group, but no significant difference for the sub-dimension of collaboration. For the sub-dimension of teaching process, it is also clear that it is at a large or larger than typical level. As a result, it can be stated that there is evidence that IEM had a positive impact on the development of self-efficacy for inclusive education. To support this finding, estimated marginal mean values of Scale 1 (see Fig. 1) and the results of Question 1 in Questionnaire 1 (see Fig. 2) can be included for comparison.

Figs. 1 and 2 provide evidence that the self-efficacy of pre-service teachers in the experimental group improved more in inclusive education. The validity of this evidence can also be examined through the opinions expressed by pre-service teachers in Questionnaire-2 (see Table 4).

Examining Table 4 reveals that pre-service teachers reported progress in the inclusion process, including approaches to students, material development, communication, etc. These aspects are considered to



**Fig. 1.** Estimated marginal means values of Scale 1 (Self-Efficacy of Teacher Training Students in Inclusive Education Environments).



**Fig. 2.** Findings of question 1 in Questionnaire-1.

**Table 4**  
Pre-service teachers' views on IEM's contribution (Sub-theme 1: Self-efficacy).

| Theme: Opinions about IEM's contribution                                                                                          | Participant codes           | f/n        |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------|
| <i>Sub-theme 1: Self-efficacy towards inclusive students</i>                                                                      |                             |            |
| I have understood inclusive education in a more qualified way and learned its application.                                        | All participants            | 11 (100%)  |
| I have learned to approach the student with special needs professionally instead of having feelings such as pity, fear, and pity. | P1, P2, P4, P5, P7, P9, P10 | 7 (63,63%) |
| I learned more about using materials with special needs students.                                                                 | P6                          | 1 (9.09%)  |
| I learned that I should act by listening to students with special needs.                                                          | P10                         | 1 (9.09%)  |

provide evidence for the development of self-efficacy in inclusive classrooms. Summarizing this finding, P10 stated:

Before this consultancy, I was not even aware that there were students with special needs in the class while planning the lesson. If I had not joined, I would probably only know this by name. I did not examine the student before, but after the meetings I observed them in detail. Beforehand, I was preparing the lesson plan generally, but I am now preparing it, taking into consideration students with special needs.

P10 emphasizes the importance of considering special needs when preparing lesson plans. Viewed holistically, it can be stated that the findings related to self-efficacy support each other. Therefore, strong and consistent evidence has been obtained regarding the contribution of inclusive education models (IEM) to pre-service teachers' self-efficacy.

### 3.2. Theme 2: development of pre-service teachers' attitudes towards individuals with special educational needs

The study examined the influence of IEM on participants' attitudes towards individuals with special educational needs through the total score obtained from Scale 2 and its sub-dimensions, i.e. liking, sensitivity or a negative attitude (see Table 5).

Table 5 demonstrates that the average scores obtained from Scale 2 and the sub-dimensions of liking, sensitivity and negative attitude in the post-test were higher in the experimental group. Table 6 shows the findings of mixed M(A)NOVA according to the values obtained from Scale 2 and the sub-dimensions.

Firstly, Scale 2 and its sub-dimensions of *liking*, *sensitivity* and *negative attitude* did not make any significant difference in the context of the pre- and post-test. Scale 2 and *liking* showed a significant difference in favour of the experimental group, but no significant differences were obtained for *sensitivity* and *negative attitude* based on group time interactions. Impact values were small for Scale 2, with *liking* found to be at a medium level. This indicates a small degree of evidence for the development of this attitude towards individuals with special educational needs in IEM-

**Table 5**  
Descriptive statistics values according to Scale 2 and Scale 2 sub-dimensions.

|                                                                              | Group        | Mean   | Std. Deviation | N  |
|------------------------------------------------------------------------------|--------------|--------|----------------|----|
| Pre-test: Scale 2 total score                                                | Experimental | 52.182 | 5.250          | 11 |
|                                                                              | Control      | 53.909 | 4.826          | 11 |
|                                                                              | Total        | 53.046 | 4.999          | 22 |
| Post-test: Scale 2 total score                                               | Experimental | 54.546 | 5.373          | 11 |
|                                                                              | Control      | 52.273 | 4.982          | 11 |
|                                                                              | Total        | 53.409 | 5.188          | 22 |
| Pre-test: The total score of the liking sub-dimension of Scale 2             | Experimental | 21.818 | 2.089          | 11 |
|                                                                              | Control      | 23.000 | 2.324          | 11 |
|                                                                              | Total        | 22.409 | 2.240          | 22 |
| Post-test: The total score of the liking sub-dimension of Scale 2            | Experimental | 23.456 | 2.162          | 11 |
|                                                                              | Control      | 22.182 | 3.093          | 11 |
|                                                                              | Total        | 22.818 | 2.684          | 22 |
| Pre-test: The total score of the sensitivity sub-dimension of Scale 2        | Experimental | 13.818 | 1.401          | 11 |
|                                                                              | Control      | 13.727 | 1.490          | 11 |
|                                                                              | Total        | 13.773 | 1.412          | 22 |
| Post-test: The total score of the sensitivity sub-dimension of Scale 2       | Experimental | 13.909 | 1.514          | 11 |
|                                                                              | Control      | 13.636 | 1.630          | 11 |
|                                                                              | Total        | 13.773 | 1.541          | 22 |
| Pre-test: The total score of the negative attitude sub-dimension of Scale 2  | Experimental | 16.546 | 2.505          | 11 |
|                                                                              | Control      | 17.182 | 2.786          | 11 |
|                                                                              | Total        | 16.867 | 2.606          | 22 |
| Post-test: The total score of the negative attitude sub-dimension of Scale 2 | Experimental | 17.181 | 2.136          | 11 |
|                                                                              | Control      | 16.455 | 2.544          | 11 |
|                                                                              | Total        | 16.818 | 2.322          | 22 |

based teaching. To support the slight development in attitudes, the estimated marginal mean values of Scale 2 (see Fig. 3) and the results of Question 2 in Questionnaire 1 (see Fig. 4) can be compared.

It can be stated that Figs. 3 and 4 provide slight evidence that the attitudes of pre-service teachers in the experimental group toward individuals with special educational needs have improved. In this context, the opinions of pre-service teachers in Questionnaire-2 (see Table 7) can also be considered.

Table 7 demonstrates evidence that pre-service teachers' awareness of students with special educational needs has increased, and their negative emotions and prejudices have decreased. This evidence aligns with findings regarding attitudes in the scale. Therefore, it can be inferred that IEM has the potential to influence pre-service teachers' attitudes toward students with special educational needs. P9's statement can be cited to support this finding. P9 stated:

Although our undergraduate theoretical course provides information about students with special needs, we do not focus on them in our teaching practice, as this tends to emphasise typically developing students. My practice has been improved by receiving feedback on weekly practices in mentoring related to inclusive education. For example, when planning the lesson, the characteristics of my students with special needs and the necessity to adapt the lesson automatically come to my mind.

During the planning process, P9 indicated a tendency towards more positive emotions, thoughts, and behaviours towards students with special educational needs. Consequently, although it is not very strong, various evidence suggesting that IEM has the potential to foster positive attitudes among pre-service teachers towards students with special educational needs has been gathered.

### 3.3. Theme 3: development of pre-service teachers' readiness for the teaching profession

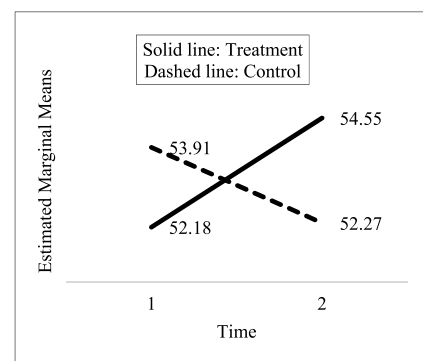
The impact of IEM on the participants' readiness for the teaching profession was examined through the total score obtained from Scale 3 and its sub-dimensions (see Table 8).

Table 8 shows that the experiment group had higher average scores obtained from Scale 3 and its sub-dimensions in the post-test. See Table 9 for the findings of the mixed M(A)NOVA analysis according to

**Table 6**  
Findings of Mixed M(A)NOVA in the context of Scale 2.

| Source                                         | SS     | df | MS     | F     | p     | $\eta^2_p$ | Power |
|------------------------------------------------|--------|----|--------|-------|-------|------------|-------|
| Between-Subjects                               |        |    |        |       |       |            |       |
| Groups                                         | .818   | 1  | .818   | .019  | .892  | .001       | .052  |
| Within-Subjects                                |        |    |        |       |       |            |       |
| Time                                           |        |    |        |       |       |            |       |
| Within Factor: Scale 2                         | 1.455  | 1  | 1.455  | .164  | .690  | .008       | .067  |
| Time*Group                                     | 44.000 | 1  | 44.000 | 4.956 | .038  | .199       | .563  |
| Between-Subjects                               |        |    |        |       |       |            |       |
| Groups                                         |        |    |        |       |       |            |       |
| The liking sub-dimension of Scale 2            | .023   | 1  | .023   | .002  | .962  | .000       | .050  |
| The sensitivity sub-dimension of Scale 2       | .364   | 1  | .364   | .097  | .759  | .005       | .060  |
| The negative attitude sub-dimension of Scale 2 | .023   | 1  | .023   | .002  | .962  | .000       | .050  |
| Within-Subjects                                |        |    |        |       |       |            |       |
| Time                                           |        |    |        |       |       |            |       |
| The liking sub-dimension of Scale 2            | 1.841  | 1  | 1.841  | .817  | .377  | 1.841      | .138  |
| The sensitivity sub-dimension of Scale 2       | .000   | 1  | .000   | .000  | 1.000 | .000       | .050  |
| The negative attitude sub-dimension of Scale 2 | .023   | 1  | .023   | .008  | .928  | .000       | .051  |
| Time*Group                                     |        |    |        |       |       |            |       |
| The liking sub-dimension of Scale 2            | 16.568 | 1  | 16.568 | 7.349 | .013  | .269       | .732  |
| The sensitivity sub-dimension of Scale 2       | .091   | 1  | .091   | .114  | .739  | .006       | .062  |
| The negative attitude sub-dimension of Scale 2 | 5.114  | 1  | 5.114  | 1.881 | .185  | .086       | .257  |

Note. SS: Sum of Squares, MS: Mean Square.



**Fig. 3.** The estimated marginal means values of Scale 2 (Readiness for the Teaching Profession).

the values obtained from Scale 3 and its sub-dimensions.

This demonstrates clear evidence for significant multivariate effects due to group and time interaction for Scale 3. The linear combinations of the related dependent variables contain evidence of a medium difference between the experimental and control groups in the context of pre- and post-test. Table 9 shows a significant difference between Scale 3 and its first sub-dimension in the pre- and post-tests in favour of the

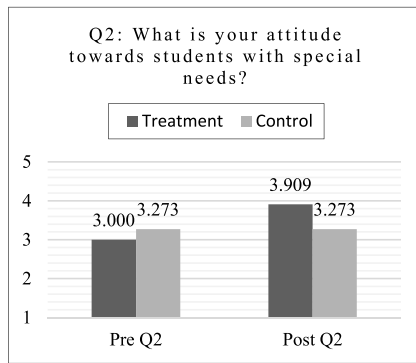


Fig. 4. Findings of question 2 in Questionnaire-1.

Table 7

Pre-service teachers' views on IEM's contribution (Sub-theme 2: Attitude).

| Theme: Opinions about IEM's contribution                                                            | Participant codes       | f/n        |
|-----------------------------------------------------------------------------------------------------|-------------------------|------------|
| <i>Sub-theme 2: Attitude towards inclusive students</i>                                             |                         |            |
| Within the scope of inclusive education, my awareness of students with special needs has increased. | All participants        | 11 (100%)  |
| My feelings of fear, sadness and anxiety towards students with special needs have decreased.        | P2, P6, P7, P8, P9, P10 | 6 (54,54%) |
| My bias for the education process with a student with special needs has ended.                      | P1, P3                  | 2 (18.18%) |

Table 8

Descriptive statistics values according to scale 3 and scale 3 sub-dimensions.

|                                                            | Group        | Mean    | Std. Deviation | N  |
|------------------------------------------------------------|--------------|---------|----------------|----|
| Pre-test: Scale 3 total score                              | Experimental | 119.273 | 10.451         | 11 |
|                                                            | Control      | 118.546 | 8.116          | 11 |
|                                                            | Total        | 118.909 | 9.139          | 22 |
| Post-test: Scale 3 total score                             | Experimental | 126.364 | 11.236         | 11 |
|                                                            | Control      | 118.546 | 10.014         | 11 |
|                                                            | Total        | 122.455 | 11.130         | 22 |
| Pre-test: The total score of 1st sub-dimension of Scale 3  | Experimental | 95.727  | 7.989          | 11 |
|                                                            | Control      | 95.091  | 6.564          | 11 |
|                                                            | Total        | 95.409  | 7.142          | 22 |
| Post-test: The total score of 1st sub-dimension of Scale 3 | Experimental | 101.727 | 8.498          | 11 |
|                                                            | Control      | 95.273  | 8.150          | 11 |
|                                                            | Total        | 98.500  | 8.771          | 22 |
| Pre-test: The total score of 2nd sub-dimension of Scale 3  | Experimental | 23.546  | 2.806          | 11 |
|                                                            | Control      | 23.455  | 2.296          | 11 |
|                                                            | Total        | 23.500  | 2.502          | 22 |
| Post-test: The total score of 2nd sub-dimension of Scale 3 | Experimental | 24.636  | 3.139          | 11 |
|                                                            | Control      | 23.273  | 2.284          | 11 |
|                                                            | Total        | 23.955  | 2.768          | 22 |

experimental group, but no difference in the second sub-dimension. This study therefore concludes that IEM-based education has a positive impact on pre-service teachers' readiness for the teaching profession. To support this positive effect, the estimated marginal mean values of Scale 3 (see Fig. 5) and the results of Question 3 in Questionnaire 1 (see Fig. 6) can be compared.

Figs. 5 and 6 provide evidence that pre-service teachers in the experimental group showed significant development in terms of readiness for the teaching profession. In this context, the opinions of pre-service teachers in Questionnaire-2 (see Table 10) can also be considered.

In Table 10, evidence shows that pre-service teachers have improved in terms of readiness for the teaching profession, specifically in professional development and personal development. They generally reported improvements in adapting lessons, gaining self-confidence, preparing active learning processes for students, and demonstrating empathy.

Table 9

Findings on Mixed M(A)NOVA in the context of Scale 3.

| Source                       | SS      | df | MS      | F     | p    | $\eta_p^2$ | Power |
|------------------------------|---------|----|---------|-------|------|------------|-------|
| Between-Subjects             |         |    |         |       |      |            |       |
| Groups                       | 200.818 | 1  | 200.818 | 1.106 | .306 | .052       | .170  |
| Within-Subjects              |         |    |         |       |      |            |       |
| Time                         |         |    |         |       |      |            |       |
| Within Factor: Scale 3       | 138.273 | 1  | 138.273 | 7.212 | .014 | .265       | .724  |
| Time*Group                   | 138.273 | 1  | 138.273 | 7.212 | .014 | .265       | .724  |
| Between-Subjects             |         |    |         |       |      |            |       |
| Groups                       |         |    |         |       |      |            |       |
| 1st sub-dimension of Scale 3 | 138.273 | 1  | 138.273 | 1.252 | .277 | .059       | .187  |
| 2nd sub-dimension of Scale 3 | 5.818   | 1  | 5.818   | .485  | .494 | .024       | .102  |
| Within-Subjects              |         |    |         |       |      |            |       |
| Time                         |         |    |         |       |      |            |       |
| 1st sub-dimension of Scale 3 | 105.091 | 1  | 105.091 | 8.550 | .008 | .299       | .794  |
| 2nd sub-dimension of Scale 3 | 2.273   | 1  | 2.273   | 1.075 | .312 | .051       | .167  |
| Time*Group                   |         |    |         |       |      |            |       |
| 1st sub-dimension of Scale 3 | 93.091  | 1  | 93.091  | 7.574 | .012 | .275       | .745  |
| 2nd sub-dimension of Scale 3 | 4.455   | 1  | 4.455   | 2.108 | .162 | .095       | .282  |

Note. SS: Sum of Squares, MS: Mean Square.

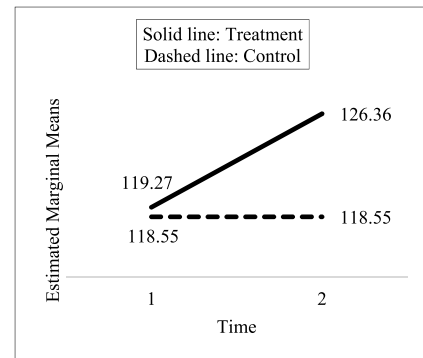


Fig. 5. Estimated marginal means values of Scale 3 (Readiness for the Teaching Profession).

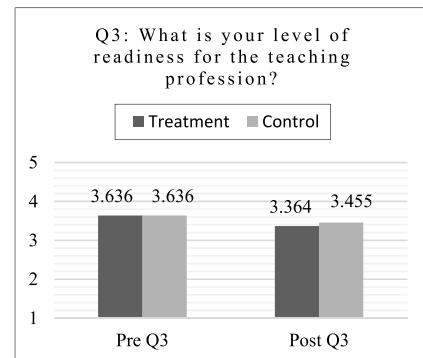


Fig. 6. Findings of question 3 in Questionnaire-1.

**Table 10**

Pre-service teachers' views on IEM's contribution (Sub-theme 3: Professional development and sub-theme 4: Personal development).

| Theme:                                                                                                                        | Participant codes           | f/n           |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------|
| Opinions about IEM's contribution: Readiness for the teaching                                                                 |                             |               |
| <i>Sub-theme 3: Professional development</i>                                                                                  |                             |               |
| I have learned to adapt the lesson.                                                                                           | All participants            | 11<br>(100%)  |
| I have learned to activate students in class.                                                                                 | All participants            | 11<br>(100%)  |
| My professional self-confidence has increased.                                                                                | P1, P2, P4, P6, P8, P9, P11 | 7<br>(63,63%) |
| It helped me use practical knowledge effectively.                                                                             | P4, P5, P6, P9              | 4<br>(36,36%) |
| It made me feel the need for individual care not only for special needs but also for every student.                           | P2, P3, P10                 | 3<br>(27,27%) |
| I have just understood better by experiencing that inclusive education also contributes to students with typical development. | P2, P4                      | 2<br>(18,18%) |
| I learned how to make environmental arrangements for students.                                                                | P9                          | 1 (9,09%)     |
| <i>Sub-theme 4: Personal development</i>                                                                                      |                             |               |
| My own social-communication skills have developed.                                                                            | P3, P4, P8, P10, P11        | 5<br>(45,45%) |
| I felt the need to do research and it allowed me to read a lot of resources and improve myself.                               | P3, P10                     | 2<br>(18,18%) |
| My empathy skills have improved.                                                                                              | P2, P11                     | 2<br>(18,18%) |
| My perspective on life has changed.                                                                                           | P7                          | 1 (9,09%)     |

They also indicated personal growth and improvement in educating both students with special educational needs and those with typical development. Thus, it is evident that these findings support each other. P2's opinion can be included to further support and summarize this conclusion.

Although we focused on individuals with special educational needs during the mentoring, I found it beneficial for all my students. For example, if I give a student the chance to choose, I have learned many things, such as that they will be more interested in the lesson. I used these with all children during the application phase.

**Table 11**

First week implementation of IEM: Problems and possible solutions for students with special needs in inclusive classrooms.

|                                                           | Main theme: Problems and suggested solutions                                                                                                                           |                                                                                                                                                     |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | Problems                                                                                                                                                               | Suggested solutions                                                                                                                                 |
| <i>Sub-theme 1: Teacher performance</i>                   | The practicing teachers lack of knowledge and skills in the inclusive classroom regarding the adaptation process.                                                      | Giving appropriate feedback<br>Giving the right reinforcement<br>Support with concrete experiences<br>Professional approach to problems<br>Teamwork |
| <i>Sub-theme 2: Student engagement</i>                    | Students get bored.<br><br>Students are reluctant in the lesson.                                                                                                       | Information about students with special needs<br>Information on individual studies and adaptations                                                  |
| <i>Sub-theme 3: Individual differences among students</i> | Students with the same diagnosis have diverse needs.<br>Students with the same diagnosis have different cognitive, social, affective, and psychomotor characteristics. | Technology supported applications<br>Games and sports supported applications                                                                        |
| <i>Sub-theme 4: Communication</i>                         | Students do not make eye contact.                                                                                                                                      | Peer support                                                                                                                                        |

P2's opinion emphasizes integration and professional development. The findings from the video recordings also support this observation. Table 11 presents the issues and potential solutions discussed by participants during the initial meeting of the IEM process.

Following the issues related to teacher performance, student engagement, individual differences among students, and communication, as well as the potential solutions indicated in Table 11, six more interactive applications were made in the context of the problems encountered by the volunteers after the initial stage within the scope of IEM. This revealed that the participants experienced difficulties in preparing content for different courses, including planning and producing solutions to problems encountered in the classroom (see Table 12).

Table 12 focuses on professional development, including the following sub-themes: (1) material development/use; (2) teaching methods; (3) lesson design; (4) feedback, correction and evaluation; (5) communication; (6) individual recognition; and (7) classroom management. The findings obtained from unstructured observations also support these findings regarding professional readiness. All those in the experimental group stated at the commencement of this process that they needed to improve their ability to deliver inclusive education. They also emphasised that the considered IEM was vital to improve these skills and to overcome the limitations of their course, which focused on theory to the expense of its school-based application. In addition, all the participants highlighted that their supervisors paid little attention to the issue of students with special educational needs or inclusive education, viewing them as lacking the relevant competences. They also considered that the teachers acting as their mentors gave them insufficient directions concerning the use of inclusive education.

Another notable element in the observation process was that all of the participants stated that IEM helped them to improve their knowledge of their profession. For example, P4 said:

We heard about inclusive education and learned about it. But frankly, I never thought about how to do it. Thanks to this mentoring, I now better understand my professional responsibilities.

#### 3.4. Theme 4: suggestions of pre-service teachers in the context of the IEM process

In questionnaire-1, an open-ended question was also asked for the only compulsory course for inclusive education in undergraduate education in the same period. In addition to all participants stating that the lesson was beneficial, their suggestions are presented in Table 13.

Table 13 presents the opinions of pre-service teachers regarding the compulsory course on special education at university. This course, being a vocational knowledge course, is expected to impact pre-service teachers' self-efficacy towards inclusive education, their attitudes towards students with special educational needs, and their readiness for the teaching profession. Upon examination of the findings at the conclusion of the IEM process, pre-service teachers suggested that the content of this course should be intensive and practice based. These suggestions may indicate variations in the perspectives of pre-service teachers regarding inclusion. To further explore these insights, the suggestions of pre-service teachers for IEM (see Table 14) were also examined.

## 4. Discussion

This study examined the use of IEM to enhance the inclusive education skills—specifically self-efficacy in inclusive education, attitudes towards individuals with special educational needs, and readiness for the teaching profession—among pre-service teachers within the framework of the experiential learning model. The data revealed that IEM was beneficial for the pre-service teachers taking part. This indicates that the results of this research are significant in relation to the training of teachers and their professional performance.



**Table 12**  
Suggestions for solving problems at IEM meetings.

|                                                                                                                                           |                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Prepared courses: Turkish, mathematics, social studies, science, visual arts, music, physical education, traffic safety, free activities. |                                                                                               |
| Main Theme: Professional Development                                                                                                      |                                                                                               |
| Sub-theme 1:<br>Material development/use (f = 12)                                                                                         | Using videos when necessary                                                                   |
|                                                                                                                                           | Making necessary adjustments to the worksheet                                                 |
|                                                                                                                                           | Paying attention to the font and length of written texts                                      |
|                                                                                                                                           | Using toys                                                                                    |
|                                                                                                                                           | Using puppets                                                                                 |
|                                                                                                                                           | Using images                                                                                  |
|                                                                                                                                           | Giving students enough time to respond                                                        |
|                                                                                                                                           | Ensuring that the materials used in the activities are not distracting                        |
|                                                                                                                                           | Selecting and developing the right materials                                                  |
|                                                                                                                                           | Utilizing tangible materials                                                                  |
| Sub-theme 2:<br>Teaching methods (f = 12)                                                                                                 | Making use of animations                                                                      |
|                                                                                                                                           | Benefiting from musical events                                                                |
|                                                                                                                                           | Using correct instructions                                                                    |
|                                                                                                                                           | Utilizing peer support                                                                        |
|                                                                                                                                           | Leveraging learning through play                                                              |
|                                                                                                                                           | Observing students in the learning process                                                    |
|                                                                                                                                           | Using experiential learning experiences                                                       |
|                                                                                                                                           | Making use of group studies                                                                   |
|                                                                                                                                           | Designing collaborative learning                                                              |
|                                                                                                                                           | Appealing to different senses                                                                 |
| Sub-theme 3:<br>Lesson design (f = 12)                                                                                                    | Using game in activities                                                                      |
|                                                                                                                                           | Making appropriate concept teaching                                                           |
|                                                                                                                                           | Giving explanatory examples for metaphorical concepts                                         |
|                                                                                                                                           | Utilizing active teaching techniques                                                          |
|                                                                                                                                           | Making quality lesson plans                                                                   |
|                                                                                                                                           | Checking the preliminary information                                                          |
|                                                                                                                                           | Ranking activities from simple to complex                                                     |
|                                                                                                                                           | Designing self-confidence building activities                                                 |
|                                                                                                                                           | Designing activities in which students can think differently                                  |
|                                                                                                                                           | Frequently revising subjects taught                                                           |
| Sub-theme 4:<br>Feedback, correction and evaluation (f = 8)                                                                               | Including motivating factors                                                                  |
|                                                                                                                                           | Making associations with daily life                                                           |
|                                                                                                                                           | Making students be active in the learning process and supporting them                         |
|                                                                                                                                           | Including experiences that will develop creativity                                            |
|                                                                                                                                           | Considering student level in the content of the activities                                    |
|                                                                                                                                           | Paying attention to student motivation                                                        |
|                                                                                                                                           | Giving constructive feedback                                                                  |
|                                                                                                                                           | Considering the student's anxiety level                                                       |
|                                                                                                                                           | Giving students enough time to respond                                                        |
|                                                                                                                                           | Being constructive in the feedback process                                                    |
| Sub-theme 5:<br>Communication (f = 7)                                                                                                     | Including right questions at the right time in learning process                               |
|                                                                                                                                           | Identifying misconceptions                                                                    |
|                                                                                                                                           | Giving homework                                                                               |
|                                                                                                                                           | Considering the student's anxiety level                                                       |
|                                                                                                                                           | Using verbal cues                                                                             |
|                                                                                                                                           | Using physical cues                                                                           |
|                                                                                                                                           | Using verbal reinforcement                                                                    |
|                                                                                                                                           | Including activities that are of a quality that will provide communication skills             |
|                                                                                                                                           | Designing entertaining activities                                                             |
|                                                                                                                                           | Using professional language                                                                   |
| Sub-theme 6:<br>Individual recognition (f = 5)                                                                                            | Diversifying clues                                                                            |
|                                                                                                                                           | Knowing extensively about students                                                            |
|                                                                                                                                           | Understanding the student's emotional state (using The Incredible 5-Point scale)              |
|                                                                                                                                           | Considering individual differences                                                            |
|                                                                                                                                           | Discovering students' interests                                                               |
|                                                                                                                                           | Observing all aspects of students' strengths and needs                                        |
| Sub-theme 7:<br>Classroom management (f = 5)                                                                                              | Working by taking a break when necessary                                                      |
|                                                                                                                                           | Correctly and frequently implementing social activities that will ensure in-class integration |
|                                                                                                                                           | Being consistent in classroom management                                                      |
|                                                                                                                                           | Matching the activities with the needs of the students correctly                              |

**Table 13**  
Pre-service teachers' suggestions for the course related to inclusive education in undergraduate education.

| Pre-service teachers' suggestions                                                                                         | Experimental group (f/n) | Control group (f/n) |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------------|
| Since the application for inclusive education is a need for undergraduate education, this course should include practice. | 11 (100%)                | 7 (63.63%)          |
| Since the course content is very intense, the course hours should be increased.                                           | 2 (18.18%)               | 3 (27.27%)          |
| The lesson was effective, I have no suggestions.                                                                          | 0 (0%)                   | 1 (9.09%)           |
| No answer                                                                                                                 | 0 (0%)                   | 3 (27.27%)          |

**Table 14**  
Pre-service teachers' recommendations for IEM.

| Suggestions of the participants                                                                             | Participant codes | f/n        |
|-------------------------------------------------------------------------------------------------------------|-------------------|------------|
| It should become a permanent practice.                                                                      | All participants  | 11 (100%)  |
| If the meeting is face-to-face, the efficiency can increase.                                                | P2, P4, P5, P8    | 4 (36,36%) |
| It should be integrated into the teacher practice process.                                                  | P6, P7            | 2 (18.18%) |
| A specialist from the field of psychological counselling and guidance may also be involved in the training. | P3                | 1 (9.09%)  |
| Our own field advisor in teaching practice can also attend the IEM                                          | P6                | 1 (9.09%)  |

The first part of this discussion examines inadequacies present in teacher training courses, as identified by the pre-service teachers at the start of IEM, and those arising from their experiences of school-based practice.

The first IEM meeting highlighted a number of important issues when it came to the training of pre-service teachers. The participants stated that they had not been adequately taught how to adapt their lesson plans according to the requirements of students with special educational needs in inclusive classes. This is significant due to this aspect being a fundamental element of inclusive education (MNE, 2018). In addition, it raises concerns that such a lack of relevant skills can have serious implications for the quality of inclusive education (Jačová et al., 2018). In this context, based on the suggestions provided by pre-service teachers, it was proposed that compulsory courses should not only be rich in content but also practice-based. This suggestion may also indicate the inadequacies of pre-service teachers as revealed within the scope of the research.

Furthermore, the pre-service teachers noted their lack of awareness of the differing characteristics of students with special educational needs (Brillante, 2017). This demonstrates an important limitation in terms of their competencies. For example, attention deficit hyperactivity disorder does not necessarily imply that pupils exhibit all associated symptoms; they may also have additional issues. In some students diagnosed with attention deficit hyperactivity disorder, attention deficit may predominate, while in others, hyperactivity may be more pronounced (Centers for Disease Control and Prevention, 2024). Although a student with attention deficit hyperactivity disorder primarily exhibits attention deficits rather than hyperactivity, it may cause problems if the teacher interprets the student based on all the symptoms of typical attention deficit hyperactivity disorder and continues the typical education. This highlights the need to consider the individual characteristics, needs and strengths of such pupils.

The above discussion highlights the inadequacies of current undergraduate education aimed at pre-service teachers, particularly the lack of any course directly related to inclusive education until the final year (Council of Higher Education [CoHE], 2018). Leaving the development

of skills such as self-efficacy in inclusive education, attitudes towards individuals with special educational needs and readiness for the teaching profession until so late in their training can be considered a serious limitation. This is due to the need for an inclusive culture (e.g. equity and diversity) to fully develop to ensure a fundamental understanding of inclusive education (UNESCO, 2017), which may take a considerable amount of time to develop. In addition, the participants criticised the focus on theory and the lack of time devoted to inclusion, particularly in relation to their school-based practice. Furthermore, this issue has been identified in many countries (European Agency for Development in Special Needs Education, 2011; CoHE, 2018), highlighting the general lack of opportunities for pre-service teachers to experience the inclusion of special needs.

Moreover, the participants in the current study also found that their supervisors tended to focus on the class in general. This is supported by a number of studies from varying countries showing the inadequacies of academics when it comes to the issue of inclusion and students with special educational needs (Author, 2023; Allam & Martin, 2021; Engelbrecht, 2020; Nikoghosyan, 2020). Thus, although teacher training differs between countries, problems relating to the quality of the educational process are experienced globally (UNESCO, 2020). In addition, the comments from pre-service teachers in this study, who represent the future teachers, indicate the potential for ongoing challenges in this area. Therefore, gaining experience with special educational needs enhances the quality of undergraduate education, allowing pre-service teachers to develop skills such as self-efficacy in inclusive education, positive attitudes towards individuals with special educational needs, and readiness for the teaching profession. Consequently, it is expected that pre-service teachers, and subsequently future educators, will enhance their proficiency in these specific areas. Therefore, it is crucial to first understand how IEM enhances self-efficacy for inclusive education and readiness for teaching.

This research demonstrates that IEM was effective in addressing certain inadequacies, such as self-efficacy for inclusive education and professional readiness for teaching in pre-service teacher training. Thus, although the control and experimental groups participated in the compulsory course and teaching practice for inclusive education in a similar manner, the experimental group showed greater improvements than the control group at the end of the mentoring period. This was seen not only in the self-efficacy of pre-service teachers for inclusive education, but also in their professional readiness for teaching. Qualitative findings also provide supporting evidence for this observed development. In this context, participants noted that pre-service teachers' knowledge, awareness, and skills regarding individuals with special educational needs had improved. Moreover, pre-service teachers indicated improvements not only in their understanding of individuals with special educational needs but also in their approach to educational processes for all students in a classroom. These outcomes suggest that the benefits attributed to IEM contribute significantly to overall readiness for the teaching profession. This is a vital observation, as the acquisition of skills related to inclusive education forms a crucial component of teachers' readiness to enter their profession (Zulfija et al., 2013).

However, it should be noted that this research found no compelling substantiation for the impact of IEM when it came to the improvement of attitudes towards students with special educational needs in either the control or the experimental groups. It may be due to the tendency for attitudes to exhibit resistance to change, necessitating a significant amount of time for any potential shifts to occur (e.g. Eagly & Chaiken, 1995; Verplanken & Orbell, 2022). The current investigation observed that, despite the similarity of the two cohorts when it came to the general attitude scale, the marked contrast between the control and experimental cohorts in the *liking* subcategory could imply that IEM may prove to be effective. Qualitative findings also support this conclusion. For instance, when examining the questionnaire results on attitudes, it was observed that the scores of the experimental group increased, unlike

those of the control group. Moreover, pre-service teachers themselves acknowledged that IEM had a positive impact on them (such as the disappearance of feelings like pity, fear, and sadness, and the reduction of prejudice). To foster further improvement, it may be necessary to implement the IEM approach over an extended period of time. This assertion is supported by a systematic literature review of studies examining students with special educational needs and inclusion, which indicates that changes in teachers' attitudes occur over time (Sahamkhadam, 2020). Therefore, there is a clear need for longitudinal research on this subject.

Overall, the current study found that IEM had considerable benefits for pre-service teachers. The participants exhibited progress in addressing the challenges they faced in inclusive classrooms and devising effective remedies. However, the study also concludes that the acquisition of fundamental concepts in inclusive education should not be exclusively limited to the IEM framework. Following the IEM process, the volunteers emphasised that some of the knowledge they had acquired overlapped with that of their existing coursework, although they acknowledged that the constraints of their course prevented a comprehensive examination of all subject matter. Moreover, as indicated by Wilson and Beard (2013), experiential learning can contribute to a profound and enduring process of education. Therefore, the use of IEM, which includes experiential learning, might offer a similar opportunity for education acquired through inclusive education in undergraduate studies.

## 5. Implications for practice

This research recognises that it is vital that every pre-service teacher should undertake the practical aspect of their course in inclusive classrooms encompassing students with recognised special needs. In addition, it recommends that the teaching practice mentoring process needs to be inclusive on the basis of Kolb's (1984) experiential learning model, particularly in light of the strong evidence for its effectiveness within the context of this research. However, it is also important that inclusive mentoring is carried out by an expert in the field of inclusion alongside the university mentor and the classroom teacher. The collaboration of this tripartite consultancy has the potential to benefit the training of pre-service teachers in school-based practice, as well as the existing teachers, particularly in inclusive education. Thus, in supporting the practical element, inclusive mentoring can also contribute to the pre-service teachers' theoretical course.

This demonstrates the benefits of the IEM-based model shown in Fig. 7. However, this research also recommends the need for more comprehensive efforts by policy makers to integrate it into the education system.

## 6. Limitations

One of the limitations of this study is that its findings may have limited generalisability, in particular due to the deliberate selection of participants and the small sample size. In addition, the lack of previous research related to IEM in the literature has led to difficulties in making comparisons, thus indicating the need for a broader experimental study. A further limitation is that, although various data collection tools were used in the research, there was a lack of access to observational data pertaining to the implementation processes of the individuals in the inclusive classrooms.

## 7. Future research directions

This study has shown that IEM can be beneficially integrated into undergraduate training for pre-service teachers. Therefore, the researcher recommends that additional studies should focus on how IEM (or similar experiential practices) can be integrated into undergraduate teacher training curriculums. The researcher also concludes that future

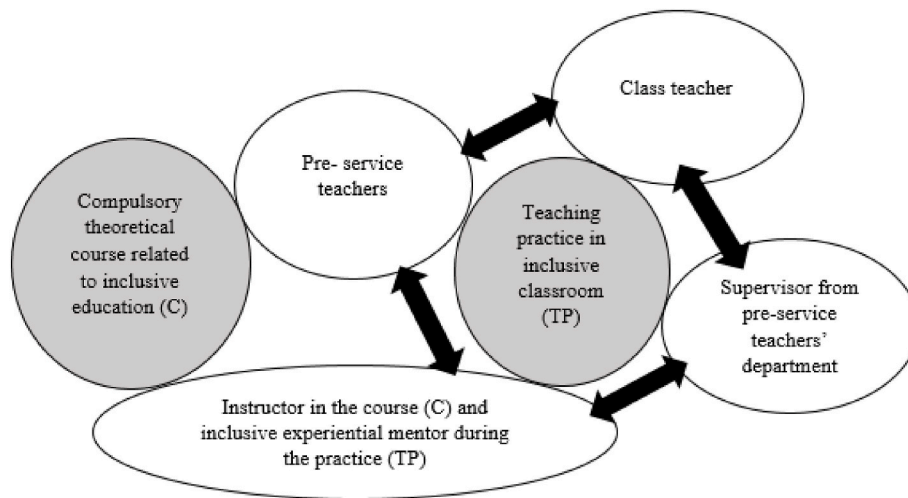


Fig. 7. IEM-based model to support pre-service teachers.

studies should focus on awareness of IEM, including its role in improving teachers' performance for all their students, in particular in inclusive classrooms. These approaches would overcome the limitations of the current study and make a significant contribution to the literature.

## 8. Conclusions

This research explored the impact of Inclusive Education Models (IEM) on pre-service teachers' development toward inclusive education. It focused on three main areas: self-efficacy in inclusive education, attitudes towards individuals with special educational needs, and readiness for the teaching profession.

Key findings indicated that participation in IEM significantly enhanced pre-service teachers' self-efficacy within inclusive classroom settings and their professional readiness. However, the study did not find substantial changes in attitudes towards individuals with special educational needs between the experimental and control groups, suggesting a need for longer-term interventions to influence attitudes effectively.

The study also highlighted deficiencies in current teacher training programs concerning the preparation of pre-service teachers to adapt lesson plans for students with special educational needs. It emphasised the importance of incorporating more practical and comprehensive coursework on inclusive education within undergraduate teacher education curricula. At this point, more longitudinal studies are needed to integrate IEM into current undergraduate education.

## CRediT authorship contribution statement

**Mahmut Serkan Yazici:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Fatma Gül Uzuner:** Writing – review & editing, Visualization, Validation, Software, Methodology, Formal analysis, Data curation.

## Declaration of competing interest

No potential conflict of interest was reported by the authors.

## Data availability

No data was used for the research described in the article.

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